A 24.46TOPS/W and 3.04TOPS/mm^2 BF16×1-bit CIM-based BERT Accelerator in 28nm CMOS

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***Abstract*—Transformer-based Large Language Models (LLMs) present severe memory and bandwidth bottlenecks on edge devices. While SRAM-based compute-in-memory (CIM) effectively minimizes data movement, existing designs face a critical trade-off between inference accuracy and system efficiency. This paper presents a fully digital SRAM-based CIM accelerator featuring algorithm-hardware co-optimization to break this trade-off. On the algorithmic front, we propose a Ternary Weight Splitting (TWS) algorithm that binarizes full-precision BERT models, achieving a 32× compression rate with negligible (0.5%) accuracy loss. On the hardware front, we propose: 1) a BF16×1-b processing element (PE) utilizing a bit-parallel SRAM macro with batch-wise mantissa alignment; 2) a Group-Vector CIM architecture capable of storing entire BERT-Tiny model columns to maximize stationary data reuse; and 3) a Data Frame Standardization scheme to eliminate tensor reshaping overhead. Fabricated in 28nm CMOS, the accelerator achieves a peak area efficiency of 3.04TOPS/mm² at 370MHz. Measurement results demonstrate energy efficiencies of 24.46TOPS/W (BF16×1-b) and 36.45TOPS/W (8-b×1-b). Compared to state-of-the-art counterparts, this work delivers 9.5× and 2.2× improvements in area and energy efficiency, respectively, enabling efficient high-accuracy LLM inference at the edge.**

***Keywords*—Compute-in-Memory, Binary, BERT, Systolic Array**

I. Introduction

The prohibitive computational and memory demands of LLMs pose significant challenges to their efficient deployment on compute- and bandwidth-constrained edge platforms. To mitigate these overheads, state-of-the-art accelerators [1-4] predominantly employ INT8 weight quantization coupled with structured sparsity. However, as illustrated in Fig. 1, this 'INT8 and Sparsity' paradigm faces a critical trade-off: non-negligible on-chip accuracy degradation induced by aggressive sparsity [4], and limited system-level energy and area efficiency caused by the intensive off-chip data movement of 8-bit weights [1-3]. Conversely, while emerging binary LLM accelerators [5] effectively minimize power consumption, their low-precision activations severely compromise inference accuracy. Consequently, adopting Brain-Floating Point-16 (BF16) activations presents a promising pathway to restore accuracy, yet it imposes three distinct design challenges on CIM architectures. First, implementing precise exponent alignment necessitates additional comparison trees and shifters, whereas coarse-grained layer-wise mantissa shifting mitigates this overhead at the expense of system accuracy [6-7]. Second, while prior CIM architectures [3] employ broadcast-based systolic scheduling to mitigate critical timing degradation caused by substantial signal fan-out, the requisite massive register insertion incurs prohibitive power and area penalties. Third, the diverse operators and varying tensor shapes in LLMs impose significant data reshaping and buffering overheads, leading to CIM under-utilization [7].

*Qijia Xie, *Weiran Chen contributed equally to this work.

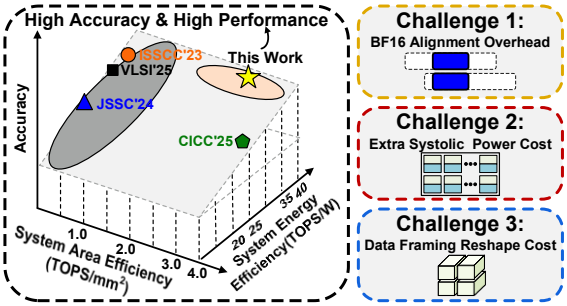


Fig. 1. The trade-off of BERT system accuracy, area efficiency, and energy efficiency.

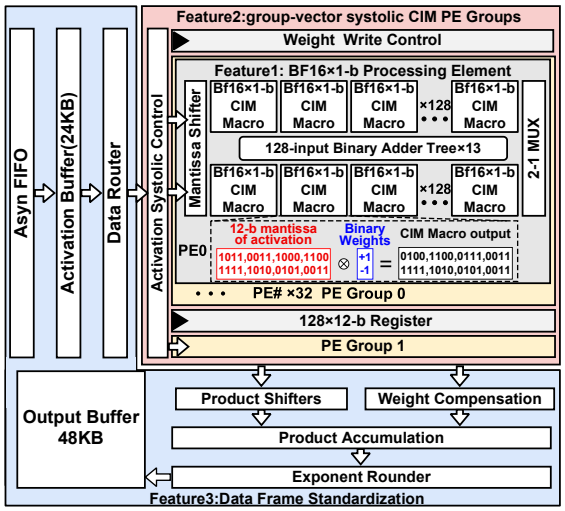


Fig. 2. The block diagram of overall BERT CIM architecture.

To address these challenges, this work presents a fully digital SRAM-based CIM accelerator featuring algorithm-hardware co-optimization to enhance both efficiency and accuracy. First, on the algorithmic front, we employ TWS binarization to obtain BF16×1-b and 8-b×1-b BERT models, achieving a 32× compression rate versus full-precision models with only 0.5% accuracy loss on the SST-2 dataset. Second, on the hardware front, we introduce a 2KB digital CIM accelerator based on a bit-parallel SRAM macro for BF16×1-b matrix multiplications. As shown in Fig. 2, the design features: (1) BF16×1-b PEs incorporating batch-wise shifting to mitigate alignment cost and

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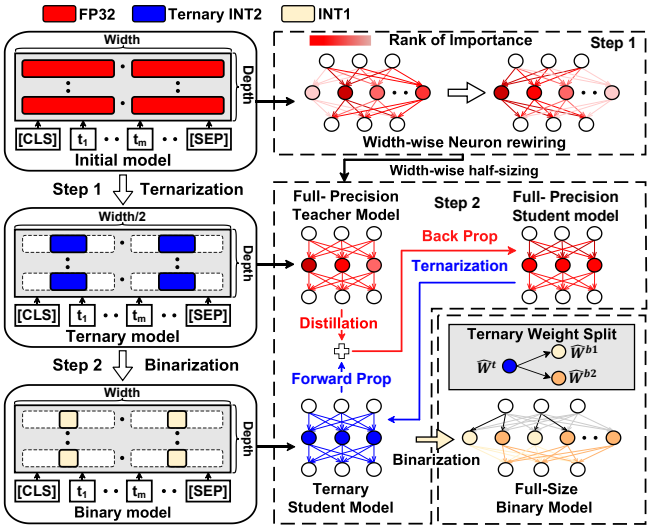


Fig. 3. The two-step binarization flow of BF16×1-b BERT model.

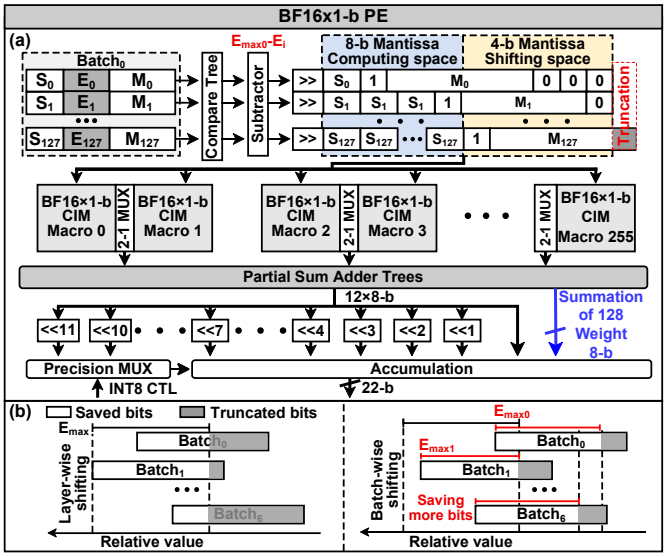


Fig. 4. The BF16×1-b CIM PE: (a) architecture and (b) comparison between layer-wise shifting and batch-wise shifting.

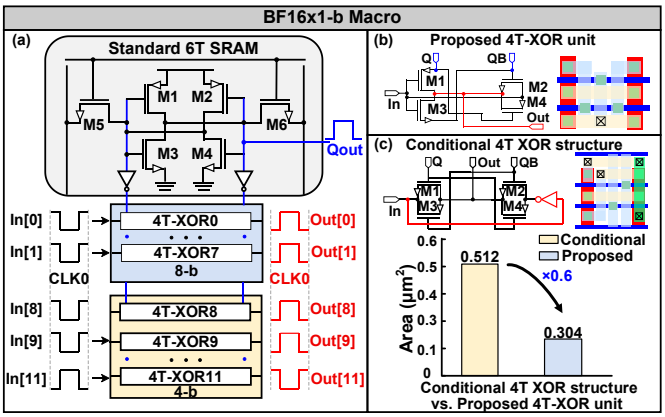


Fig. 5. The BF16×1-b CIM macro: (a) proposed schematic, (b) proposed layout and (c) comparison with conditional 4T XOR structure.

balance accuracy trade-offs; (2) Group-Vector PE groups with a synchronous weight update mechanism to simultaneously maximize throughput and energy efficiency; and (3) standardized data frames to eliminate reshaping overhead and buffering latency. Measurement results demonstrate 9.5× and 2.2× improvements in area and energy efficiency over the state-of-the-art counterparts, marking the first silicon-verified CIM accelerator for binary transformers to simultaneously break the trade-off among area, energy, and accuracy.

II. BERT Binarization

Directly training binary transformers is non-trivial due to the steep and irregular loss landscape, which often results in significant accuracy degradation. To address this, we propose a coarse-to-fine two-step binarization framework, illustrated in Fig. 3, which utilizes ternary weights as a proxy to bridge the precision gap. To further mitigate quantization error, Knowledge Distillation (KD) with a full-precision teacher is employed throughout the binarization process.

Step 1: Ternary Quantization with BF16 Activations. Unlike prior quantization methods [8] relying on INT8/4 activations, this work integrates BF16 activation quantization to improve accuracy while enhancing CIM computational efficiency. To bypass the optimization difficulty of binary weights, a half-width ternary model is first trained as a stable proxy. In this phase, full-precision weights W are quantized into ternary values $\widehat{W}$ using a layer-specific threshold Δ^l.

Step 2: Ternary Weight Splitting. In this phase, TWS is introduced to initialize the full-width binary model by splitting the trained half-width ternary weights. Mathematically, TWS decomposes each ternary weight $\widehat{W}$ into two binary components, $\widehat{W}^{b1}$ and $\widehat{W}^{b2}$, thereby doubling the channel width to restore the original model dimension. This process approximates the latent full-precision weights as a linear combination of binary bases with optimized scaling factors, effectively converting operations into streamlined BF16×1-b computations.

III. CIM Accelerator Architecture

A. BF16×1-b CIM PE

As illustrated in Fig. 4, the proposed CIM PE integrates 256 BF16×1-b CIM macros and time-multiplexes a set of thirteen 128-input binary adder trees via 128 2-to-1 multiplexers. To facilitate efficient exponent pre-alignment, a batch-wise mantissa shifter with a 4-bit extension is incorporated. In contrast to conventional layer-wise approaches, this scheme mitigates bit redundancy by aligning mantissas to the local maximum exponent (E_{max}) within each batch. Consequently, the PE executes 12-bit mantissa MAC, utilizing 12-b integer MAC operations [9]. It also supports 8-b×1-b integer operations through fine-grained gating of the shifter and adder trees.

The detailed schematic of the proposed BF16×1-b macro is illustrated in Fig. 5. It comprises a standard 6T SRAM bit-cell for 1-bit weight storage coupled with 12 mantissa partial product units implemented via 4T-XOR gates together forming a 58T SRAM-based macro. By leveraging weight sharing and XOR duplication, this bit-parallel architecture achieves single-cycle execution, representing a 12× latency reduction compared to bit-serial counterparts. Furthermore, the proposed 4T-XOR topology exploits the differential Q and QB SRAM outputs to eliminate the threshold voltage drop in typical transmission-gate structure, while achieving 40% area and 13% power reductions compared to inverter-based implementations.

B. Group-Vector Systolic PE Array

Building upon the Vector Systolic Array (VSA) concept [10], we propose a Group-Vector Systolic Array tailored for CIM to enhance

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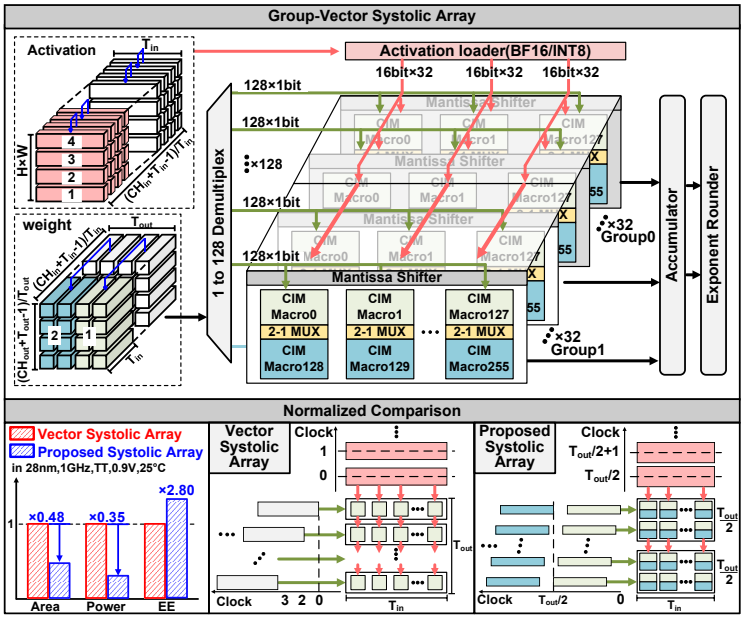


Fig. 6. The proposed group vector systolic array for CIM architecture.

parallelism via optimized register usage without timing degradation. As illustrated in Fig. 6, the array comprises two PE groups, each integrating 32 PEs with a total storage capacity of 128×128 1-bit weights. It operates under a weight-stationary dataflow where 128 input activations are broadcast within each group per cycle and systolically propagated across groups. Partial sums from each PE are processed by a dedicated exponent rounder that performs batch-wise alignment and truncates results to the BF16 format. To mitigate the timing degradation caused by massive parallelism, weight loading is restricted to the PE level. The resulting write latency is effectively masked by a synchronous weight update mechanism that concurrently updates one macro row while another performs computation. This architecture significantly outperforms conditional atomic arrays and vector arrays. Specifically, for a matrix multiplication with a T_{in}×T_{out} weight tile, the proposed architecture (with 2 groups) reduces the cycle count to T_{out}+1, in contrast to 2T_{in}+T_{out}-1 and T_{in}+T_{out}-1 required by atomic and vector arrays, respectively. Furthermore, by reducing inter-PE register overhead by 96%, the proposed array achieves area and power reductions of 51% and 65%, respectively, without compromising peak throughput under identical simulation conditions.

C. Data Frame Standardization

To maximize group-vector parallelism and eliminate under-utilized cycles, a unified data packaging strategy based on channel-packing is implemented. Departing from conventional mappings [7], this scheme aggregates adjacent channels into dense compute blocks, standardizing all operands into a 4D format: ((CH_{in}+T_{in}-1)/T_{in}, H, W, T_{in}). The blocking parameters are rigorously coupled to the hardware constraints, where T_{in}=128 aligns with the array's input parallelism and T_{out}=64 matches the physical PE array size. This standardized frame exhibits intrinsic extensibility for variable-dimension operators. For low-dimensional tensors such as LayerNorm, the tensor is ((CH_{in}+T_{in}-1)/T_{in}, Emb, T_{in}), where the embedding dimension (Emb) is spatially mapped to H and W. Conversely, high-dimensional tensors in Multi-Head Self-Attention (MHSA) are reorganized into ((CH_{in}+T_{in}-1)/T_{in}, Head×H, W, T_{in}), effectively folding the head dimension. Crucially, this alignment ensures that the output tensor format structurally mirrors the input requirement of subsequent layers. This seamless I/O consistency

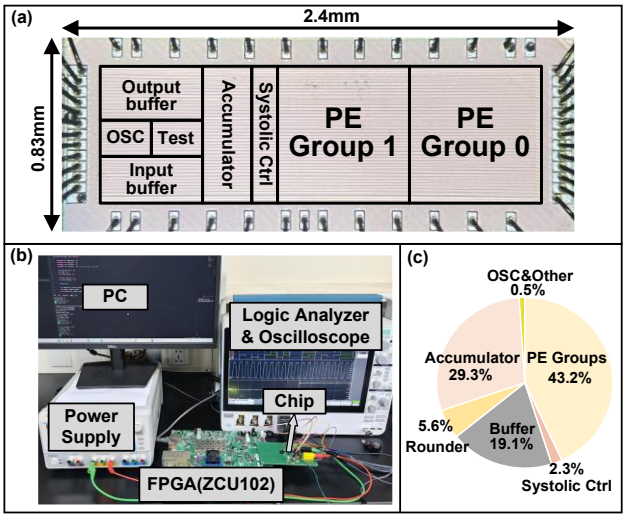


Fig. 7. (a) Die photograph, (b) test platform and (c) area breakdown.

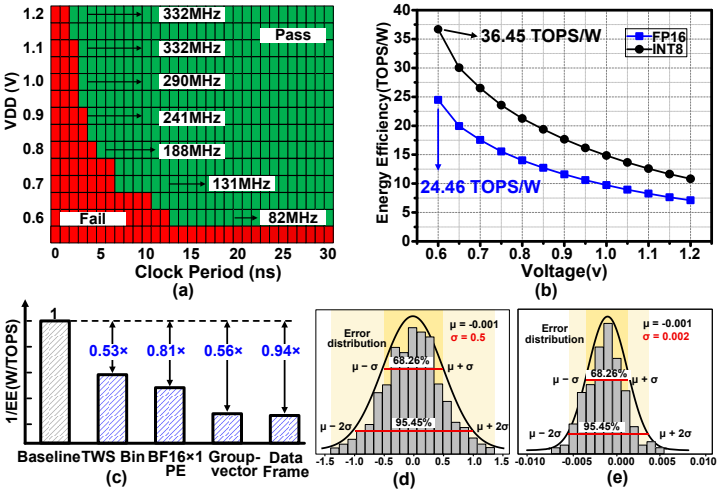


Fig. 8. Measurement results: (a) shmoo plot of the proposed CIM accelerator in BF16×1b mode. (b) Energy efficiency varies with voltage. (c) Performance evaluation. Floating point computational accuracy comparison between (d) the conventional CIM and (e) the proposed CIM.

guarantees that the group-vector array operates at peak occupancy without stalling for data realignment.

IV. Results

A. Chip Design and Measurements

The proposed accelerator is fabricated in 28nm CMOS technology, occupying a die area of 1.99 mm² with a 2KB CIM array and 72KB on-chip SRAM. Fig. 7 presents the die micrograph and detailed area breakdown, along with the measurement setup utilizing a ZCU102 FPGA for control generation and non-linear operations. As shown in the breakdown, the CIM PE groups dominate the layout, accounting for 43.2% of the total area, while the systolic control logic is highly optimized, consuming only 2.3% of area. Fig. 8 summarizes the key measurement results. The test chip operates across a frequency range of 82-370MHz with a supply voltage scaling from 0.6V to 1.2V. It achieves a peak throughput of 6.06TOPS at 1.2 V, corresponding to a state-of-the-art area efficiency of 3.04TOPS/mm². At 0.6V, the accelerator consumes 54.92mW and 38.32mW for BF16×1-b and 8-b×1-b modes, yielding peak energy efficiencies of 24.46TOPS/W and 36.45TOPS/W, respectively. This demonstrates a 4.4× improvement

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TABLE I. MEASUREMENT RESULT AND COMPARISON WITH THE STATE-OF-THE-ART TRANSFORMER ACCELERATOR

Design	ISSCC'23 [1]	VLSI'25 [2]	JSSC'24 [3]	CICC'25[4]	This work		
Technology	12nm	22nm	28nm	22nm	**28nm**		
Implementation	N.A	N.A	Digital CIM	Digital CIM	**Digital CIM**		
Precision(Bits)	A×W=FP4/8	A×W=4/8-b	A×W=8-b×8-b	A×W=8/16-b	**A×W=BF16/8-b×1-b**		
CIM Size(KB)	N.A	N.A	24	72	**2**		
On-chip SRAM (KB)	647	192	192	144	**72**		
Die Area(mm^2)	4.60	5.80	6.98	6.58	**1.99**		
Supply Voltage(V)	0.62~1.00	0.55 – 0.90	0.60-1.00	0.55-1.00	**0.60-1.20**		
Frequency (MHz)	77~717	75-400	80~275	84-536	**82~370**		
Power(mW)	9.0~122.0	19.6-197.1	20.4~119.7	-	**36.9-852.5**		
Throughput (TOPS)	0.73(FP4) 0.37(FP8)	1.74(4-b)[*1] 0.92(8-b)[*1]	2.24(8-b) 0.56(16-b)	31.60(4-b)[*1]	**1.34-6.06(BF16)**		
Area Efficiency ($TOPS/mm^2$)	0.16(FP4) 0.08(FP8)	0.30(4-b) 0.15(8-b)	0.32(8-b) 0.08(16-b)	4.80(4-b)	**3.04(BF16)**		
Model structure	BERT-Base	BERT-Base	BERT-Tiny	BERT-Base	BinaryBERT (BERT-Base)		BinaryBERT (BERT-Tiny)
Dataset	SST-2	SST-2	SQuAD 2.0	GLUE(SST-2)	SST-2	SQuAD 1.1/2.0	SQuAD 2.0
Accuracy Score	90.7%(FP4) 92.2%(FP8)	88.6%(4-b) 90.8%(8-b)	47.81%(16-b)	80.6%(8-b)	**91.5%(8-b)** **92.4%(BF16)**	**88.3%/76.5%(8-b)** **88.7%/76.7%(BF16)**	**46.5%(8-b)** **46.8%(BF16)**
Normalized EE[*2] (TOPS/W)	8.55 (FP4) 3.77 (FP8)	16.56(4-b) 8.27 (8-b)	15.71[*4]	16.77[*4]	**30.44(8-b)** **21.94(BF16)**	**32.21/29.02(8-b)** **22.36/20.88(BF16)**	**36.45(8-b)** **24.46(BF16)**
FOM[*3]	1.37	4.97	1.25	80.49	**74.35(BF16)110.81(8-b)**		

*1:System Peak *2:Normalized to 28nm at 0.65V operational power (Reported EE×(Node/28nm)×(Voltage2/0.6V^2)) *3:FOM=Normalized Energy Efficiency ×Area Efficiency *4：include nonlinear operations

over the vector-based BF16×4-b baseline. Specifically, binary quantization contributes a 1.9× efficiency gain by significantly reducing the adder count, while the group-vector architecture delivers a 1.8× improvement by minimizing register overhead. Finally, enhanced PE macro-parallelism offsets shifter and rounder overheads, netting a 1.2× efficiency boost. Furthermore, the proposed batch-wise mantissa shifter suppresses computational error by two orders of magnitude, trading an approximate 5% area and power overhead for significant accuracy restoration.

B. System Performance Evaluation

System performance is evaluated focusing on binarized BERT accuracy and benchmarking against state-of-the-art designs. As demonstrated in Table I, the proposed accelerator improves BERT deployment accuracy by 1.7%–3.8% on datasets such as SST-2, outperforming prior transformer accelerators [1-4]. Benefiting from the TWS quantization algorithm, the BF16×1 configuration incurs a negligible 0.5% accuracy drop on the SST-2 dataset, while delivering a 0.5%–2.0% accuracy gain and a 4×–8× higher compression ratio over INT4/8 quantization [8]. Furthermore, enabled by precise batch-wise alignment, the proposed CIM accelerator suffers only a 0.3% on-chip accuracy drop, significantly lower than sparse designs [4]. In terms of energy efficiency, the design achieves normalized metrics of 24.46TOPS/W and 36.45TOPS/W respectively on BF16×1-b and 8-b×1-b configuration, representing a 1.8×-8.1× improvement over prior arts. Regarding area efficiency, leveraging the bit-parallel architecture and optimized timing enables the CIM array to achieve a 34.5% higher operating frequency compared to existing designs [3], while reducing latency by 4×. Consequently, this yields up to 10.8× and 38× improvements in throughput and area efficiency, respectively. In terms of the Figure-of-Merit (FoM) considering both area and energy efficiency, this work demonstrates a comprehensive advantage of 1.4×-88.6× over state-of-the-art counterparts.

V. CONCLUSION

This paper presents a fully digital compute-in-memory accelerator for efficient binary BERT inference on edge devices, successfully breaking the inherent trade-off among inference accuracy, energy efficiency, and area efficiency. By employing TWS and KD, the proposed BF16×1-b configuration reduces the model size to 32% of the full-precision BERT-Base while limiting the accuracy drop to a negligible 0.5% on the SST-2 dataset. To maximize hardware efficiency, we introduce a highly parallel BF16×1-b CIM macro and a group-vector systolic PE array. Furthermore, a batch-wise mantissa shifter and exponent rounder are integrated to preserve computational precision. Measurement results demonstrate that the accelerator, fabricated in 28nm CMOS, achieves an area efficiency of 3.04TOPS/mm², alongside state-of-the-art energy efficiencies of 24.46TOPS/W and 36.45TOPS/W for BF16×1-b and 8-b×1-b modes, respectively. Benchmarking against state-of-the-art designs, this work delivers up to 38× and 8.1× improvements in area and energy efficiency, respectively, while maintaining the highest on-chip inference accuracy.

ACKNOWLEDGMENT

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